

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

April 2013

CL209 Structural Analysis-I

Date: 24.04.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

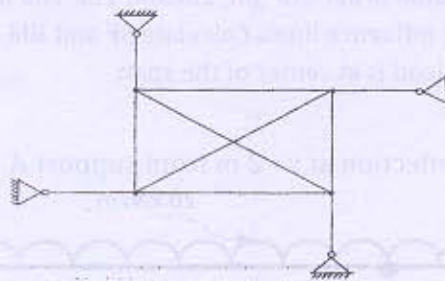
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1** A cylindrical shell, 100 mm internal diameter, wall thickness 6mm, 950mm long, is filled with an incompressible fluid at atmospheric pressure. If additional 12000 mm³ is pumped into the cylinder. Calculate the pressure exerted by the fluid on the wall of the cylinder and circumferential stress. Take $E = 200\text{GPa}$ and $\mu = 0.3$ [07]

- Q - 2 (a)** Determine Degrees of Kinematic Indeterminacy for frame given in Figure. [04]

(a) considering axial deformation (b) axial deformation of the members is neglected



- (b) Attempt any Two** [10]

- (i) A short hollow column of external and internal diameters 200 mm and 150 mm respectively carries an eccentric load 40kN. Find the greatest eccentricity at which the load cannot develop any tension at the base.
- (ii) A column of T section having flange of 200 mm x 20 mm and web of 15 mm x 180mm is subjected to a load of 100kN at a point on centroidal axis, 40mm below the centroidal X-X axis. Calculate the maximum and minimum stress induced in section.
- (iii) The dimensions of a masonry pillar are 8m high and 3m x 5m in section. A horizontal wind pressure of 1200 N/mm² acts on the face 3m x 8m. Find the maximum and minimum stress intensities at the base. Take unit weight of masonry is 20,000N/mm³.

- Q - 3** Attempt any Two [14]

- (i) A Solid round bar 4m long and 6 cm in diameter is used as a strut. Determine the crippling load for the following conditions, (a) One end is fixed and other end is free (b) Both ends are fixed (c) One end is fixed and other is hinged. Take $E = 2 \times 10^5 \text{ N/mm}^2$.
- (ii) Determine the crippling load for a T- section of dimensions 12 cm x 2 cm flange and 2 cm x 12 cm with length of 6 cm when it is used as a strut with both ends hinged. Take $E = 2 \times 10^5 \text{ N/mm}^2$.

- (iii) A column of circular section has 150 mm diameter and 3m length. Both ends of the column are fixed. The column carries a load of 100kN at an eccentricity of 15 mm from the geometrical axis of the column. Find the maximum compressive stress on the column section. Find also the maximum permissible eccentricity to avoid tension in the column section. Take $E = 1 \times 10^5 \text{ N/mm}^2$.

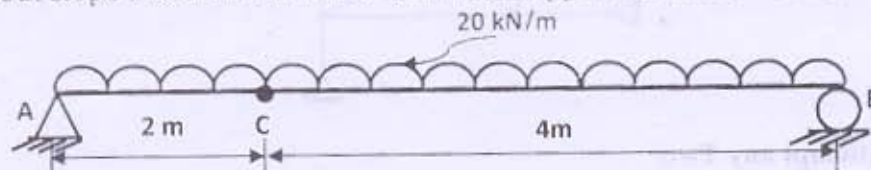
SECTION – II

- Q - 4 A three hinged parabolic arch rib has a span of 90 m and rise of 20 m to the central pin at the crown. The rib carries load of intensity 3 kN/m over a length of $1/3$ of the span from the left hand hinge. Calculate the B.M. at the quarter span points. [07]
- Q - 5 (a) Enlist advantages of three hinge arch. [02]
- (b) Attempt any Two [12]

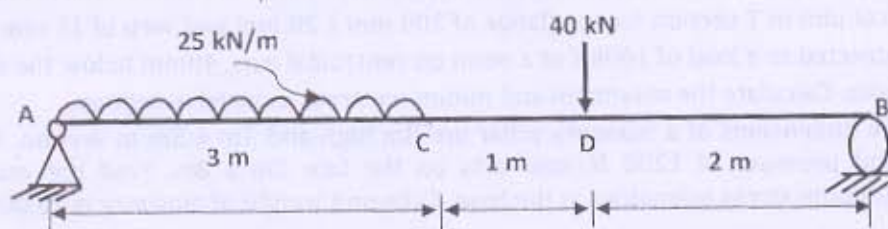
- (i) A simply supported girder has a span 30m. Six wheel loads 50kN each, spaced 2m apart cross the girder from left to right. Find maximum BM at section 8m from the left hand support.
- (ii) Find maximum SF and BM at section 4m from left end of simply supported beam of span 10m carrying UDL of 12kN/m on 3m length crossing beam from left to right.
- (iii) Four wheel loads of 6, 4, 8 and 5 kN cross a girder of 20m span from left to right followed by U.D.L of 4kN/m and 4m long with the 6kN load leading. The spacing between the loads in same order are 3m, 2m and 2m. The head of the udl is at 1m from the last 5kN load using influence lines. Calculate SF and BM at a section 8m from the left support when the 4kN load is at center of the span.

- Q - 6 Attempt any Two [14]

- (i) Find out slope and deflection at $x = 2 \text{ m}$ from support A. Take $E = 6 \times 10^4 \text{ kN.m}^2$.



- (ii) By using Macaulay's method find slope at A and deflection at D of a beam as shown in Figure. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 3 \times 10^8 \text{ mm}^4$.



- (iii) Determine deflection at C and slope at D for a simply supported beam as shown in Figure using conjugate beam method. Take $I = 20 \times 10^9 \text{ mm}^4$ & $E = 200 \text{ GPa}$.

